Project Name

FAKE NEWS DETECTION SYSTEM

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Project Report – UML & System Analysis

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WORKBOOK TEMPLATE

Project Name: Fake News Detection System

1. Problem Statement

- Fake news spreads quickly through social media, websites and messaging platforms.
- Users may not be able to verify whether a news article is genuine or misleading.
- The system analyzes submitted news text and predicts whether it is likely to be real or fake.
- The system presents the prediction result and basic supporting information to the user.

2. Why Does the Problem Exist?

- Rapid online sharing
- Lack of source verification
- Misleading headlines and edited content
- Limited media literacy
- Information overload
- Difficulty checking multiple sources

3. Internal Stakeholders

- System Administrator – manages users, models and system records.
- ML/Model Administrator – monitors model performance and training data.
- System Developer – maintains the application, database and prediction service.

4. External Stakeholders

- Users/Readers – submit or paste news for checking.
- News/Content Sources – provide article text or source information.
- Educational/Research Institutions – may use results for awareness and study.

5. Expected System Users

- Administrator
- Registered User
- General Reader

WORKBOOK – OBJECTIVES & OUTPUTS

6. Objectives of the Project

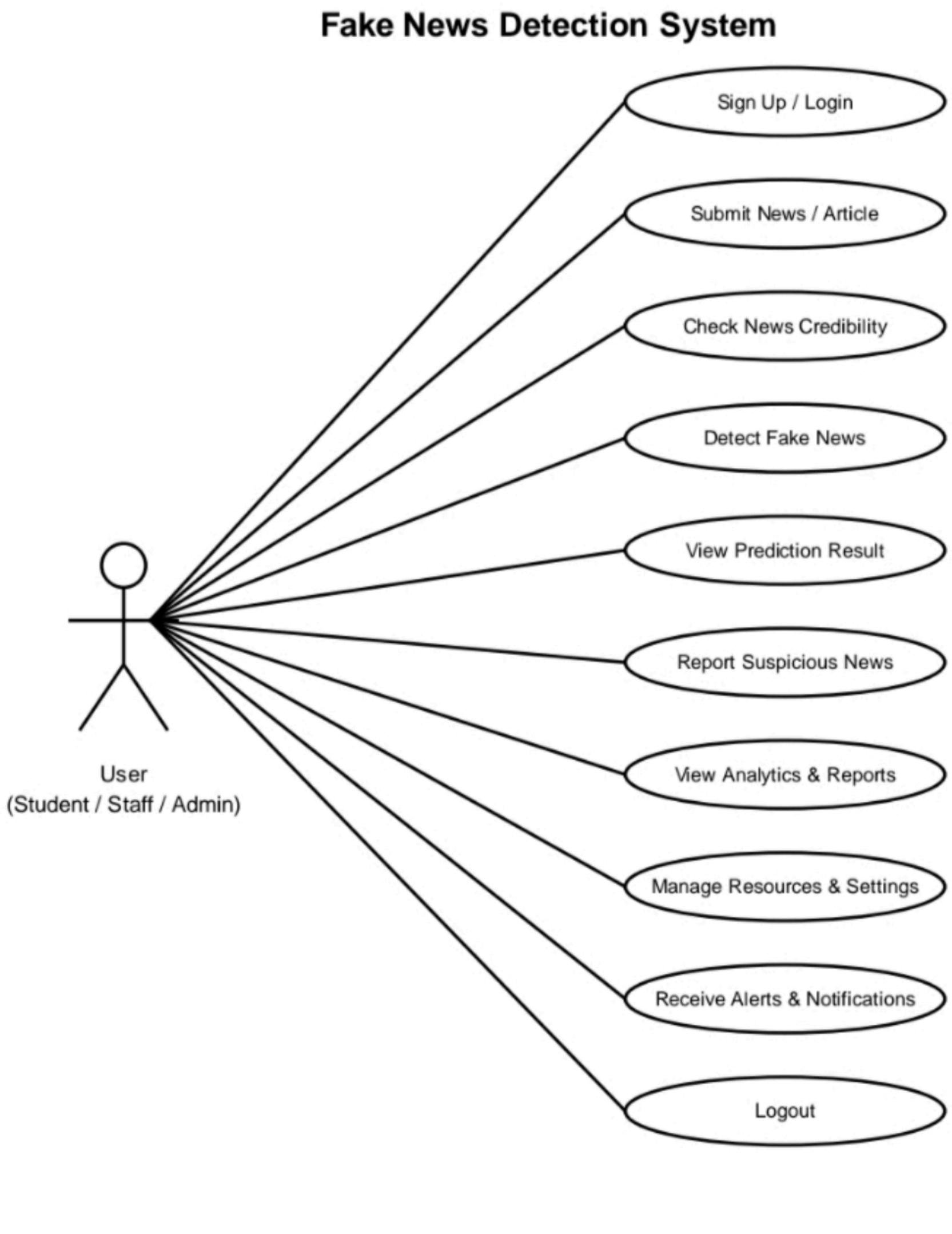
- Develop an online Fake News Detection System.
- Allow users to enter or paste news content for analysis.
- Extract useful text features and send them to a trained classification model.
- Predict whether submitted content is likely to be Real or Fake.
- Store prediction history and generate basic reports for the administrator.

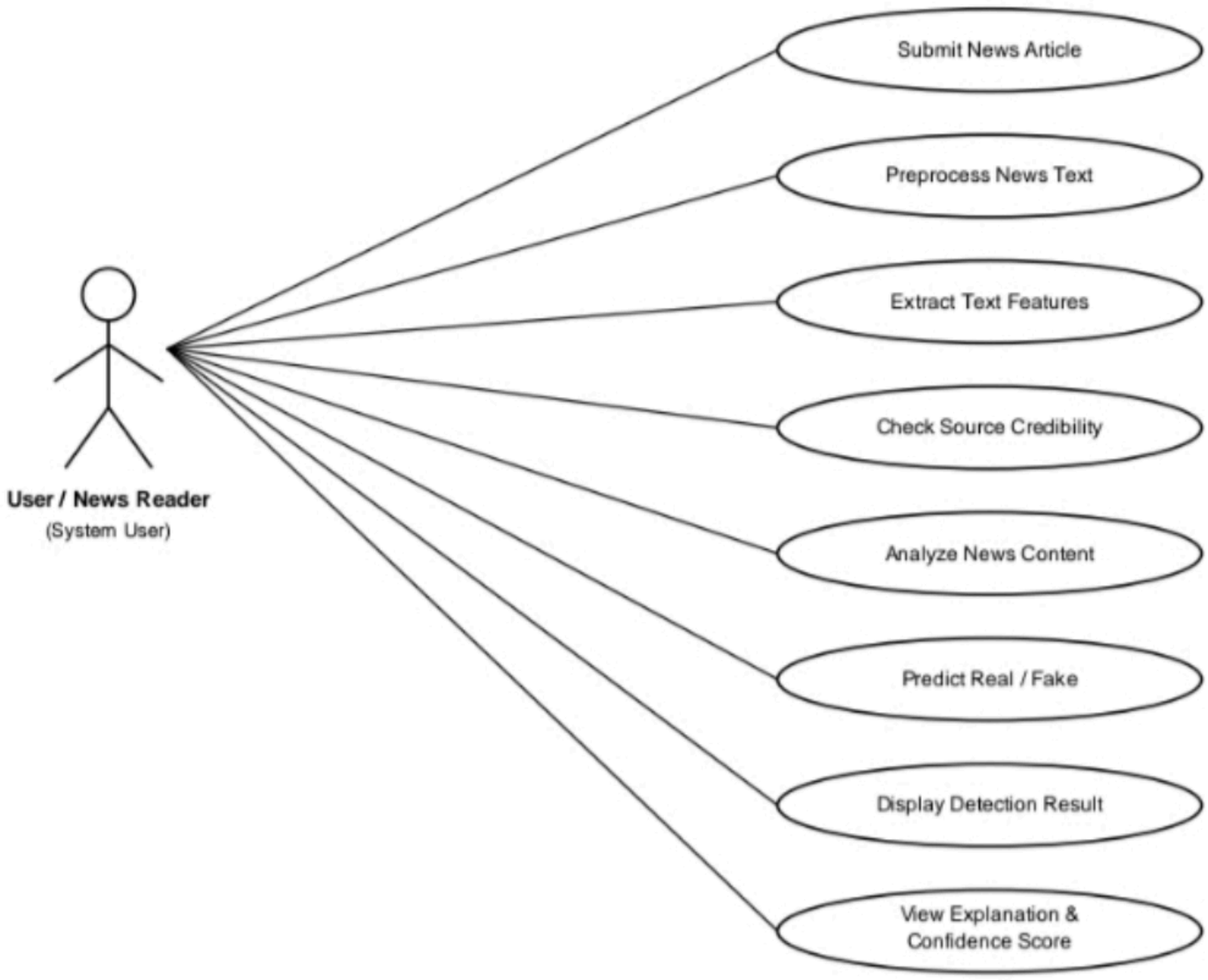
7. Expected Outputs

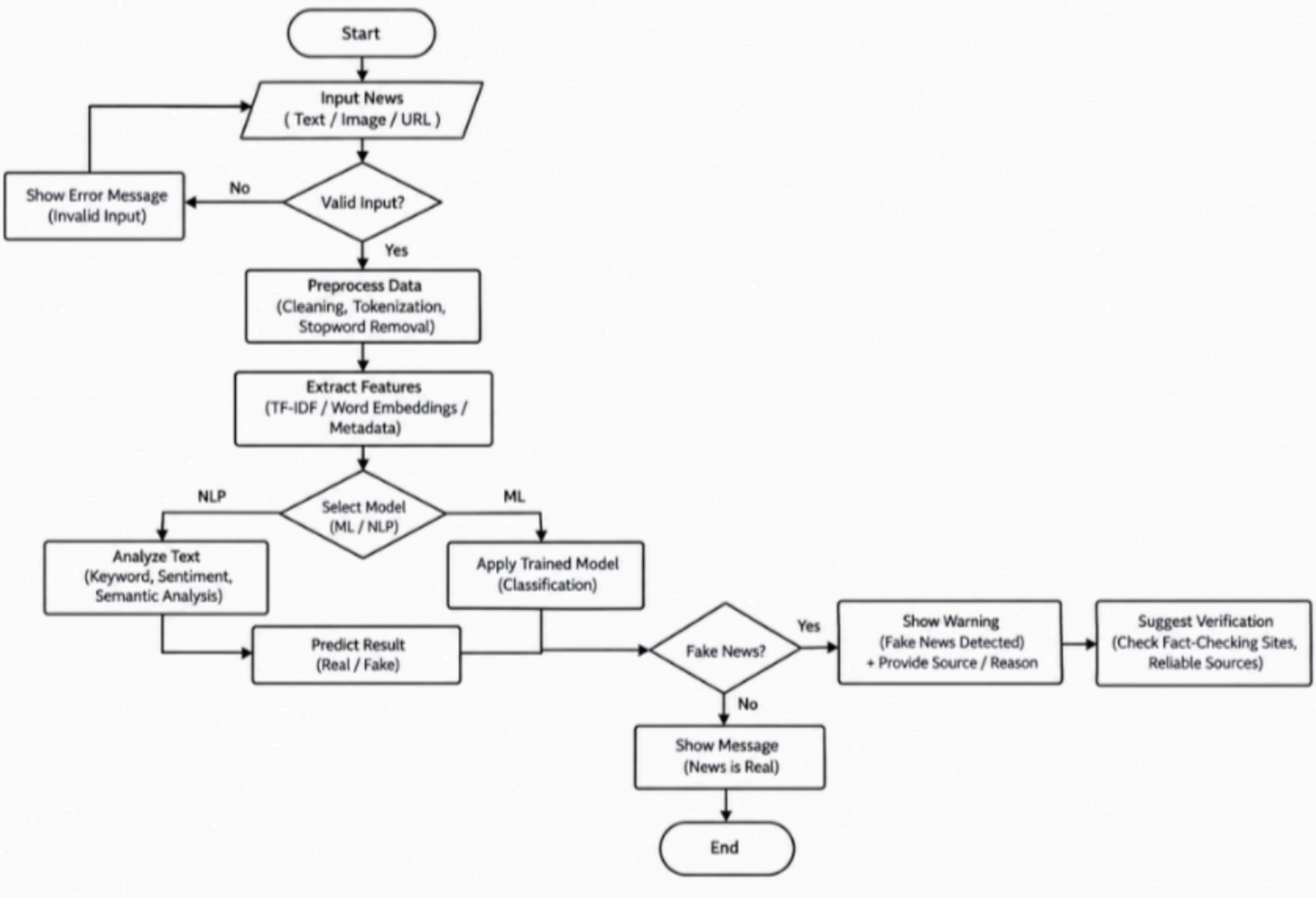
- News input form
- Prediction result
- Confidence/probability score
- Prediction history
- Admin dashboard
- Model and dataset records
- Basic reports

8. Hands-on Activity

- Collect and label a suitable news dataset.
- Clean and preprocess text data.
- Convert text into machine-readable features.
- Train and test a classification model.
- Design the website interface and prediction workflow.
- Design database tables for users, news records, predictions and model information.

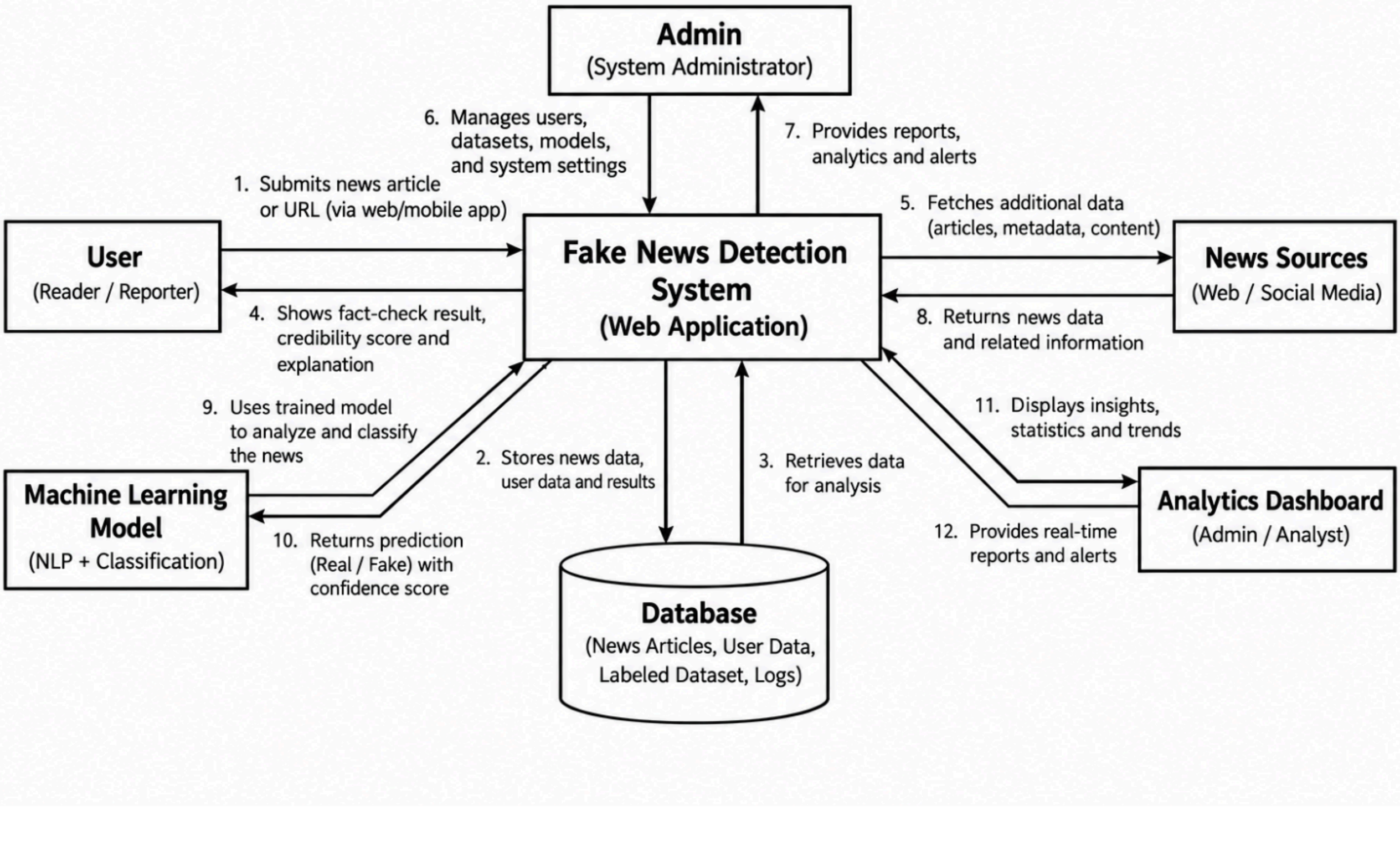

Fake News Detection System
Sign Up / Login
Submit News / Article
Check News Credibility
Detect Fake News
View Prediction Result
Report Suspicious News
View Analytics & Reports
Manage Resources & Settings
Receive Alerts & Notifications
Logout
User
(Student / Staff / Admin)


Submit News Article
Preprocess News Text
Extract Text Features
Check Source Credibility
Analyze News Content
Predict Real / Fake
Display Detection Result
View Explanation &
Confidence Score
User / News Reader
(System User)


Start
Input News
(Text / Image / URL)
Valid Input?
No
Show Error Message
(Invalid Input)
Yes
Preprocess Data
(Cleaning, Tokenization,
Stopword Removal)
Extract Features
(TF-IDF / Word Embeddings /
Metadata)
Select Model
(ML / NLP)
NLP
ML
Analyze Text
(Keyword, Sentiment,
Semantic Analysis)
Apply Trained Model
(Classification)
Predict Result
(Real / Fake)
Fake News?
Yes
Show Warning
(Fake News Detected)
+ Provide Source / Reason
Suggest Verification
(Check Fact-Checking Sites,
Reliable Sources)
No
Show Message
(News is Real)
End

Fake News Detection System

(Collaboration Diagram)



Fake News Detection System
(Class Diagram)

